

UNIT #2

Corrosion

→ Introduction ⇒ Most of the metals occur in combined state in the nature. The pure metals, in isolated form, has higher energy & it is thermodynamically unstable.

→ Metal $\xrightleftharpoons[\text{Metallurgy}]{\text{Corrosion}}$ Metallic compound + Energy.

Thus all metals tends to go back to its combined state.

This process is known as corrosion: "Any process of deterioration of a metal through an unwanted chemical or electrochemical attack by the environment is called corrosion".

→ Theories of Corrosion

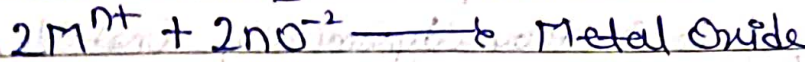
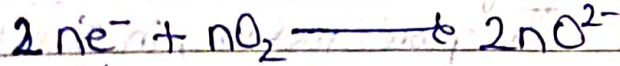
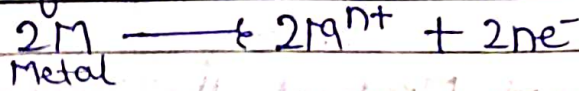
There are 3 basic theories of corrosion.

- ① Dry or chemical corrosion.
- ② Wet or electrochemical corrosion.
- ③ The Acid theory.

→ ① Dry or chemical corrosion ⇒ The dry corrosion occurs through the direct chemical attack of gases like O_2 , H_2 , H_2S , N_2 , SO_2 , Cl_2 or anhydrous liquids on the surface of metals etc.

② Corrosion by oxygen (Oxidation corrosion) ⇒ In the absence of moisture, oxygen attack on the metals to form oxides. Alkali metals & alkaline earth metals are highly reactive towards oxygen even at room temperature. Other metals (except Ag, Au & Pt) are oxidised at higher temperatures.

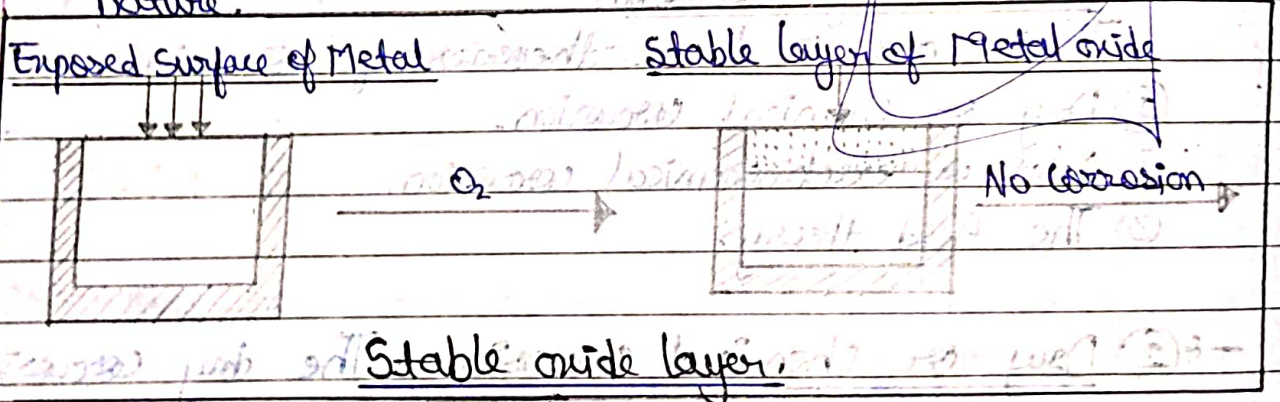
* Mechanism of oxidation (corrosion)



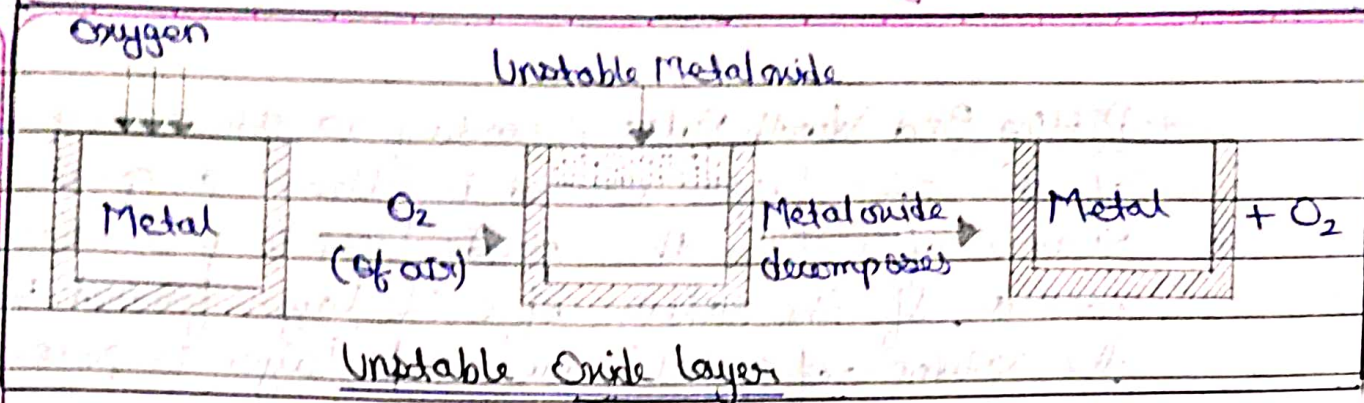
The nature of metal oxide decides the further corrosion reactions.

* Nature of Metal Oxide

- (i) Stable:- The stable metal oxide is formed by heavy metals. The layer formed behaves like a protective coating. eg: Cu, Al, etc. forms oxide layer which is protective in nature.

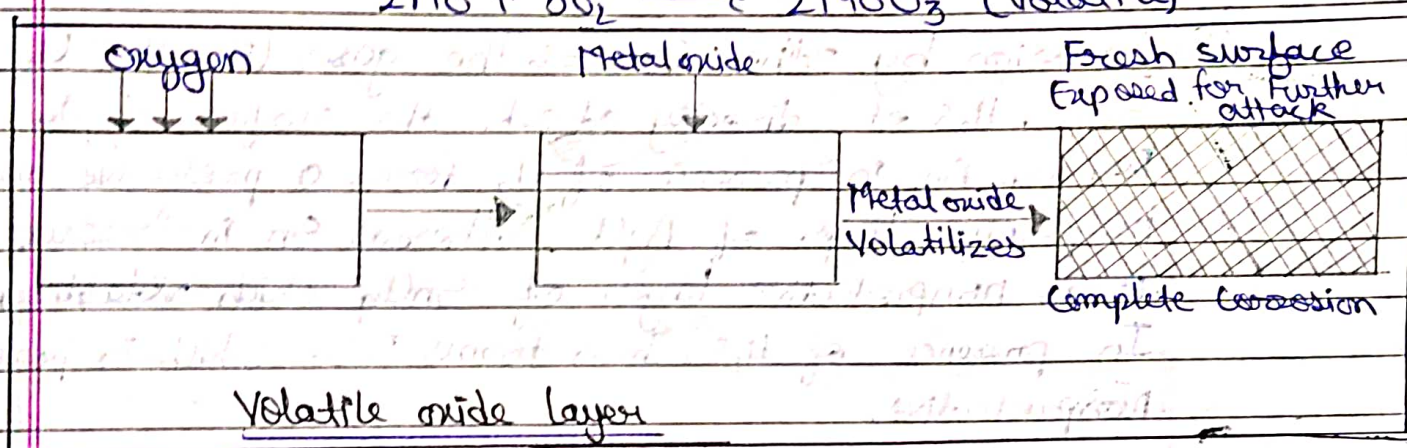
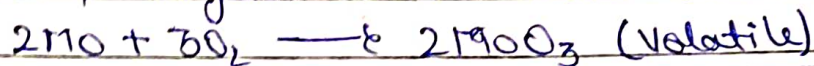


- (ii) Unstable:- When oxide layer formed is unstable, it decomposes back to metal & oxygen i.e. a temporary oxidation only takes place. Hence corrosion does not take place. e.g. Au, Ag, Pt, etc. forms unstable oxide layer.
- Metal oxide $\rightleftharpoons$ Metal + oxygen



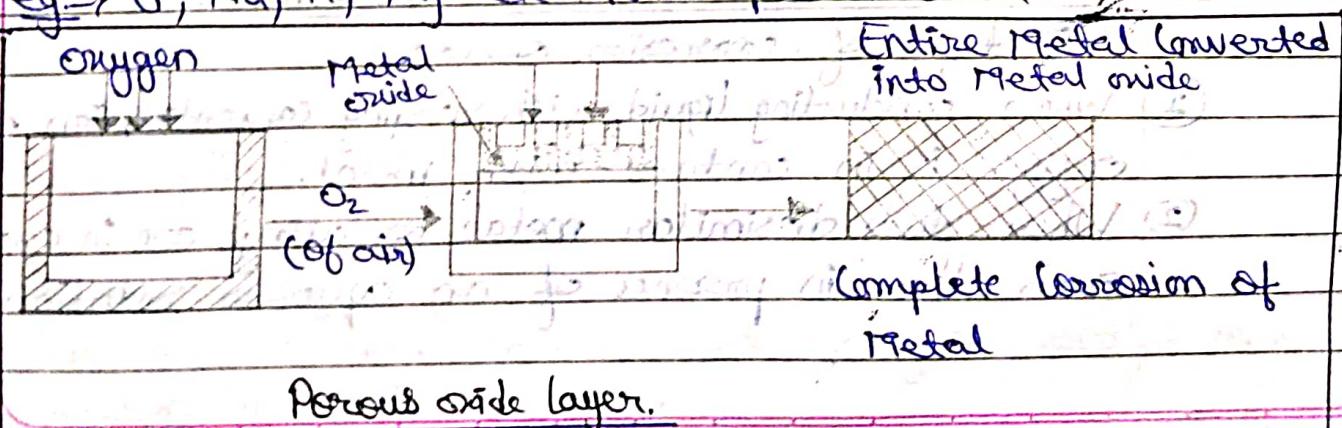
(iii) Volatile ⇒ The oxide layer volatilizes as soon as it is formed. So this causes rapid & continuous till the metal disappears.

eg ⇒ Molybdenum & tungsten forms volatile oxides.



(iv) Porous ⇒ When metal oxide formed has pores or cracks, the atmospheric oxygen penetrates inside the metal through these pores & cracks & corrosion till the entire metal is converted into oxides.

eg ⇒ Li, Na, K, Mg etc. forms porous oxides.



- (i) Volume of oxide layer $>$ Volume of metal $\Rightarrow$ oxide layer is Protective & Non porous.
 (ii) Volume of oxide layer $<$ Volume of metal $\Rightarrow$ oxide layer is Non-protective & porous.

Pitting Bed Worth Rule: According to this rule, if the volume of the oxide layer formed is atleast as great as the volume of metal then oxide layer is protective or non porous. If the volume of oxide layer formed is less than the volume of metal, the oxide layer is porous or non protective.

Thus, Li, Na, K etc. forms layer of porous oxides of less volume of metals & is therefore non protective.

Whereas, metal like Al forms a protective layer of oxide whose volume is higher than the volume of metal.

(b) Corrosion by other gases $\Rightarrow$ The gases like H_2 , Cl_2 , CO_2 , SO_2 , H_2S etc. directly attack the surface of the metals. For eg. Ag in presence of Cl_2 forms a protective and nonporous layer of $AgCl$. Whereas Sn in presence of Cl_2 forms nonprotective layer of $SnCl_4$ which volatilizes off. In presence of H_2S , Iron forms FeS , which is porous & nonprotective.

(c) Anhydrous liquids $\Rightarrow$ Anhydrous liquids, like phenols, Picric acid (organic liquids); HF , Br_2 (inorganic liquids) and liquid metals attacks the metal surface directly & causes corrosion.

(2) Wet or Electrochemical Corrosion :-

This type of corrosion occurs,

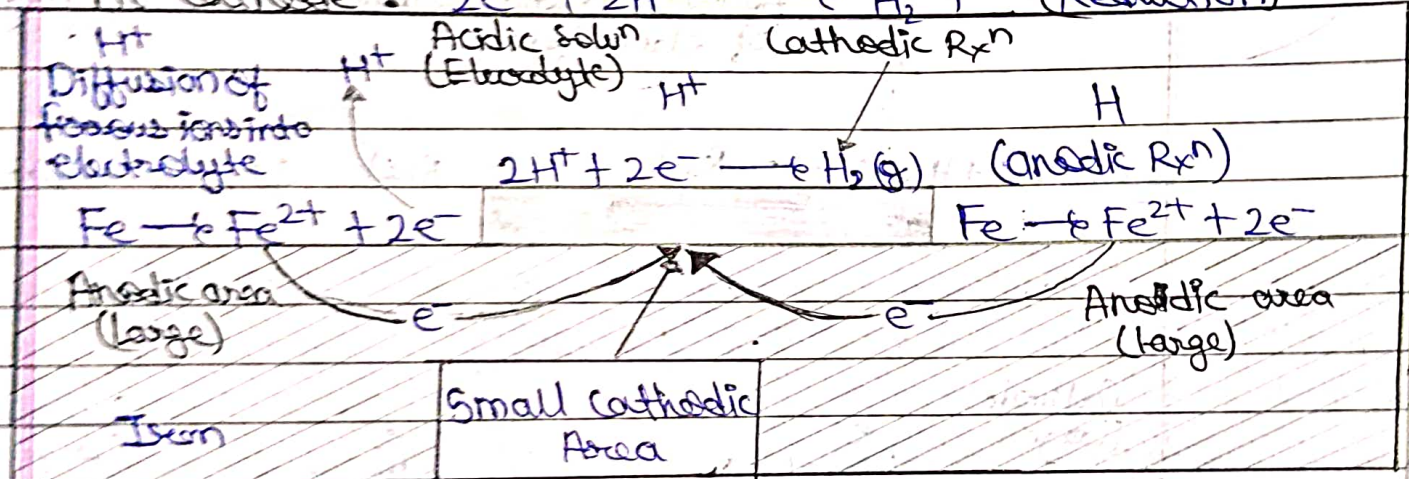
- (1) When conducting liquid with varying concentration of oxygen is in contact with metal.
- (2) When two dissimilar metals or alloys are in contact with each other in presence of an aqueous solution.

* Mechanism:- The corrosion of metal in aqueous solution occurs in two steps. Oxidation & reduction. Oxidation takes place at anode, whereas reduction takes place at cathode. The cathodic reaction may be either by evolution of hydrogen or absorption of oxygen.

* Evolution of Hydrogen:- This type of corrosion occurs, usually, in acidic medium. Metal displaces the hydrogen ions from the acidic solution with evolution of H_2 .

At anode : $Fe \longrightarrow Fe^{2+} + 2e^-$ (Oxidation)

At Cathode : $2e^- + 2H^+ \longrightarrow H_2 \uparrow$ (Reduction)



Mechanism of wet corrosion by hydrogen evolution

In hydrogen evolution type of corrosion usually the anode has large area, whereas cathode has small area.

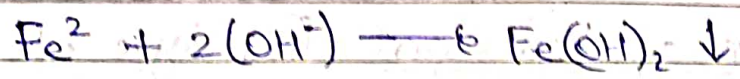
All metals above hydrogen in the electrochemical series have a tendency to get dissolved in acidic solutions ~~and~~ releasing hydrogen.

* Absorption of Oxygen.

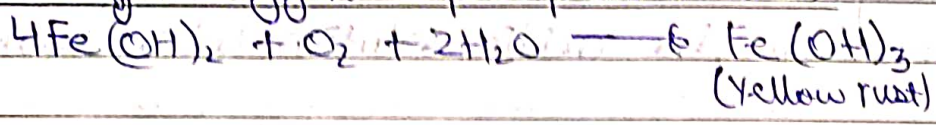
This type of corrosion occurs generally in neutral aqueous solution like NaCl solution.

- At anode : $\text{Fe} \longrightarrow \text{Fe}^{2+} + 2\text{e}^-$ (Oxidation)
- At Cathode : $\frac{1}{2}\text{O}_2 + \text{H}_2\text{O} + 2\text{e}^- \longrightarrow 2(\text{OH})^-$ (Reduction)

→ The Fe^{2+} & $(\text{OH})^-$ ions diffuse towards each other (smaller Fe^{2+} diffuse more rapidly) and form $\text{Fe}(\text{OH})_2$.

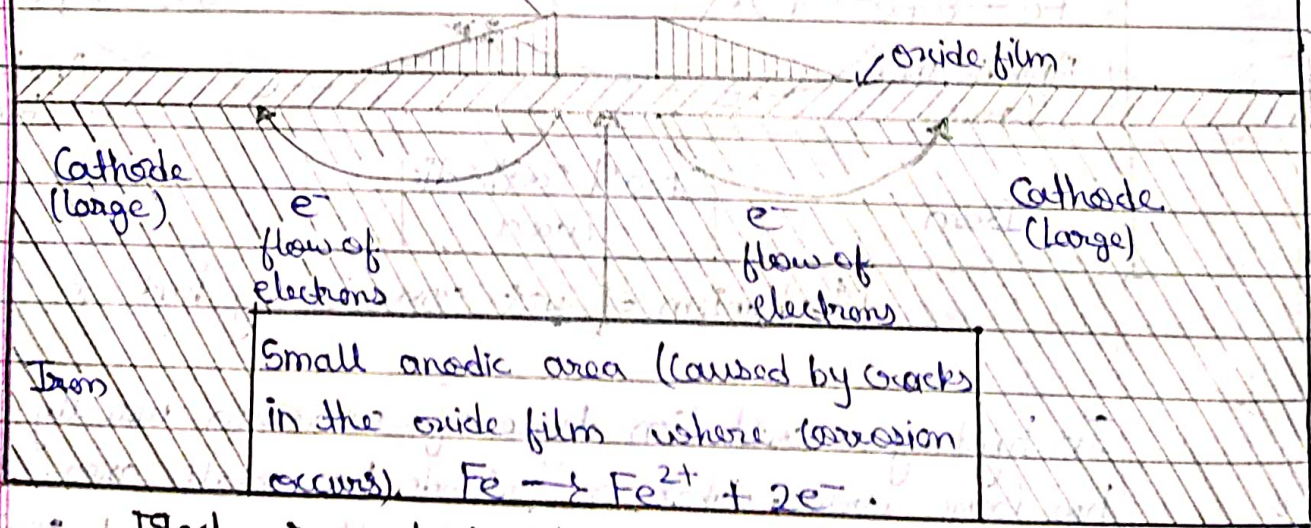
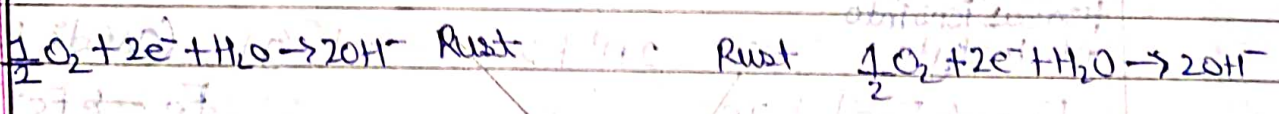


→ If enough oxygen is present then



→ If oxygen is less, then it forms black Fe_3O_4 (anhydrous magnetite)

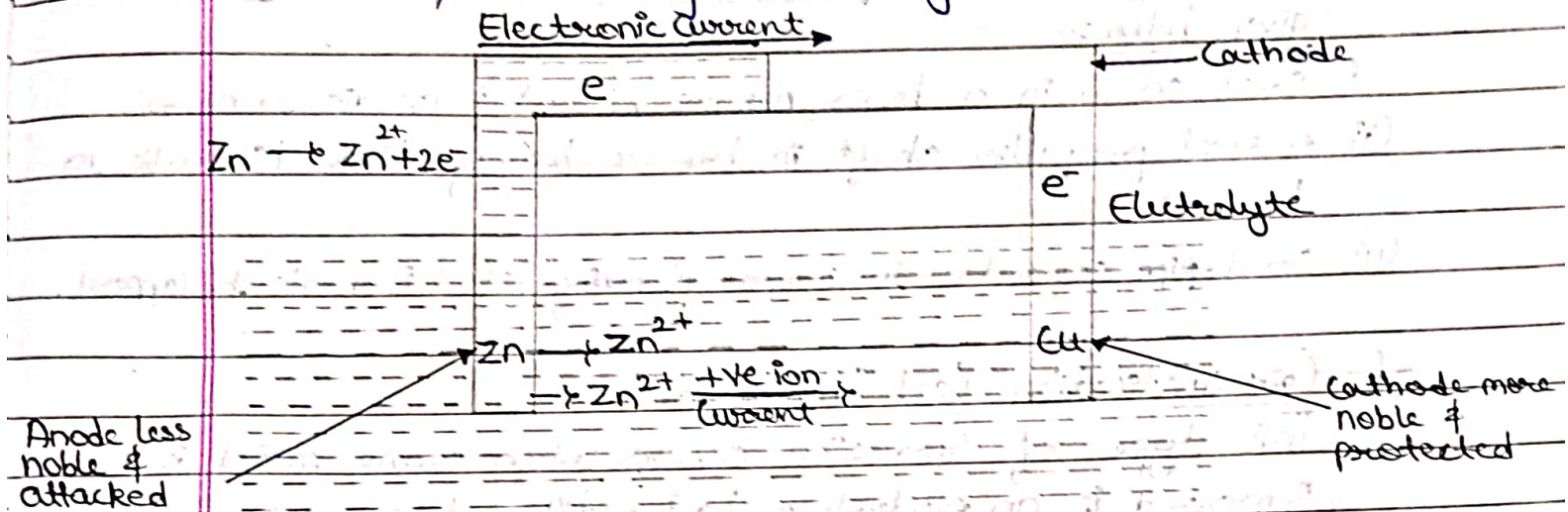
Aqueous neutral solution
of electrolyte



Mechanism of Wet Corrosion by oxygen absorption

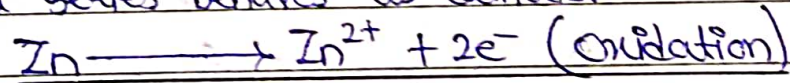
- In oxygen absorption type of corrosion, the anodic areas are small, while the cathodic area are large.

* Galvanic Corrosion → When two dissimilar metals are immersed in an electrolyte solution, and are electrically connected, then they form a galvanic cell.



Galvanic Corrosion

The metal higher in the electrochemical series becomes anode & tends to be attacked. For eg When Zn & Copper are electrically connected & immersed in an electrolyte, then the Zn which is higher in electrochemical series acts as anode & undergoes corrosion. Whereas copper which is lower in electrochemical series behaves as cathode.



At cathode, in acidic solution corrosion takes place by hydrogen evolution process, whereas, in neutral aqueous solution oxygen absorption mechanism occurs.

Other examples of galvanic couples are Zn & Ag, Fe and Cu etc.

* The Rate of galvanic corrosion depends upon the relative areas of cathode & anode. A small anodic area will be attacked faster compared to a larger one.

Rate of attack & area of cathode area of anode

→ Some of the examples where galvanic corrosion occurs, are given below:-

- (i) Steel screw in a brass machine. (Steel is anodic to brass)
- (ii) A steel propeller shaft in bronze bearing. (Steel is anodic to bronze)
- (iii) Steel pipe connected to copper plumbing (steel is anodic to copper).

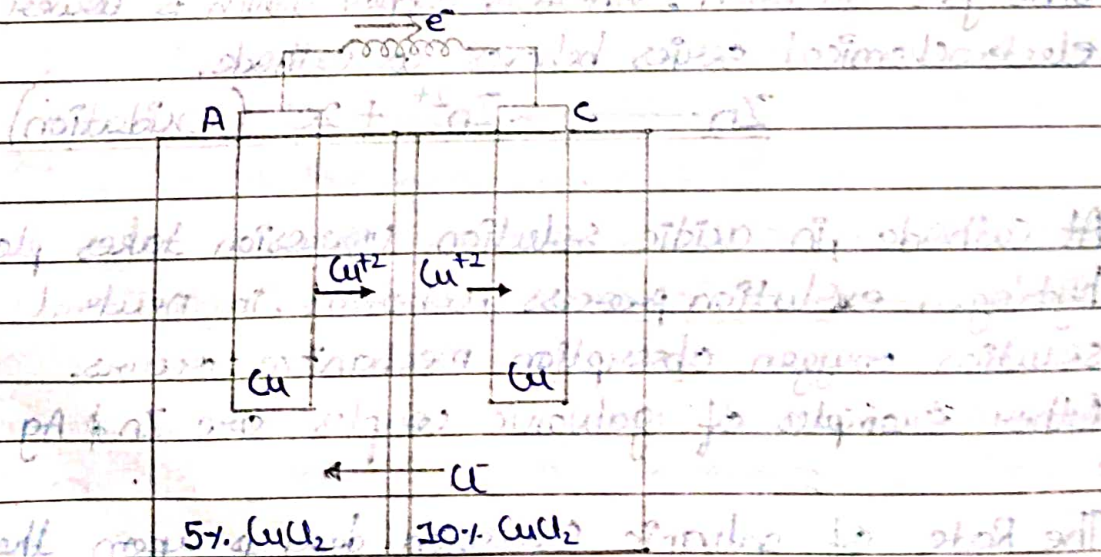
Concentration Cell Corrosion

This type of corrosion occurs when same metal is immersed in an electrolyte having different concentrations or different concentrations of dissolved oxygen. The difference in concn develop a difference in potential. The flow of current is known as different current.

Some of the examples of Concentration cell corrosion are:-

- (i) If a copper rod is in contact with one dilute and one concentration of copper salt solution.

Metal Ion Concentration Cell

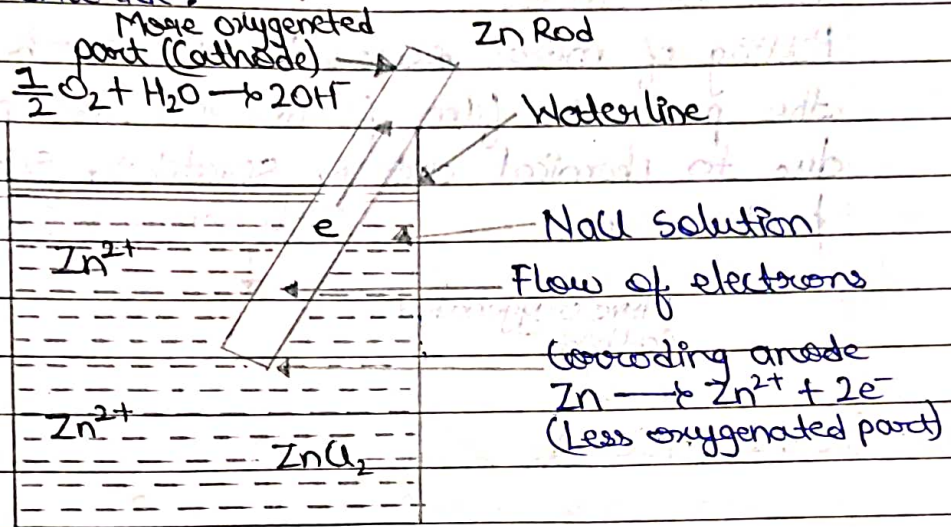


Metal Ion Concentration Cell

The part in contact with dilute solution acts as anode & will be corroded.

~~(ii) The part in contact~~

(ii) Consider a metal (Zn) partially immersed in a dilute soln of sodium chloride.



Oxygen Concentration Corrosion

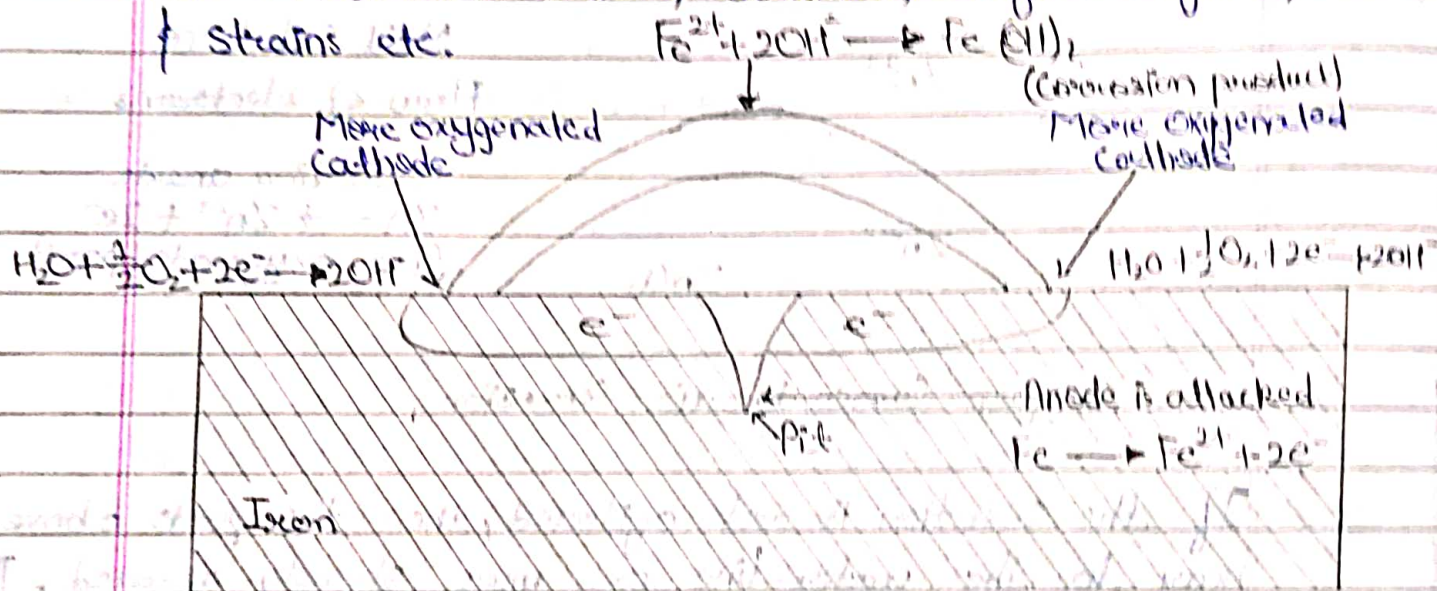
If the solution is not agitated, then the parts above & near to the water line are more strongly aerated. This part which is having a higher concentration of oxygen act as cathode. The immersed part of metal in lower concentration of oxygen becomes anode.

- At anode $\Rightarrow Zn \rightarrow Zn^{2+} + 2e^-$ (oxidation)
- At Cathode $\Rightarrow \frac{1}{2} O_2 + H_2O + 2e^- \rightarrow 2(OH^-)$ (reduction)

~~Pitting Corrosion~~ Pitting corrosion is one of the most severe, localised & accelerated form of corrosion. It may be due to formation of galvanic or concentration cells. It results in the formation of pits, cavities, & pin holes which penetrates deeply through the metal. Pitting may require a long initiation period.

If the pits are small, they may be corrosion products. So it is difficult to find the pitting corrosion till a sudden service failure occurs. The pit will act as anode. Since the area of anode is small a very high rate of corrosion will be there.

Pitting of metal occurs due to break down or cracking in the protective film formed on the metal. It may be due to chemical attack, scratches, surface roughness, stress & strains etc.



A pit is initiated by attacking anions like chloride ions.

At anode: $\text{M} \rightarrow \text{M}^{2+} + 2\text{e}^-$ (Metal)

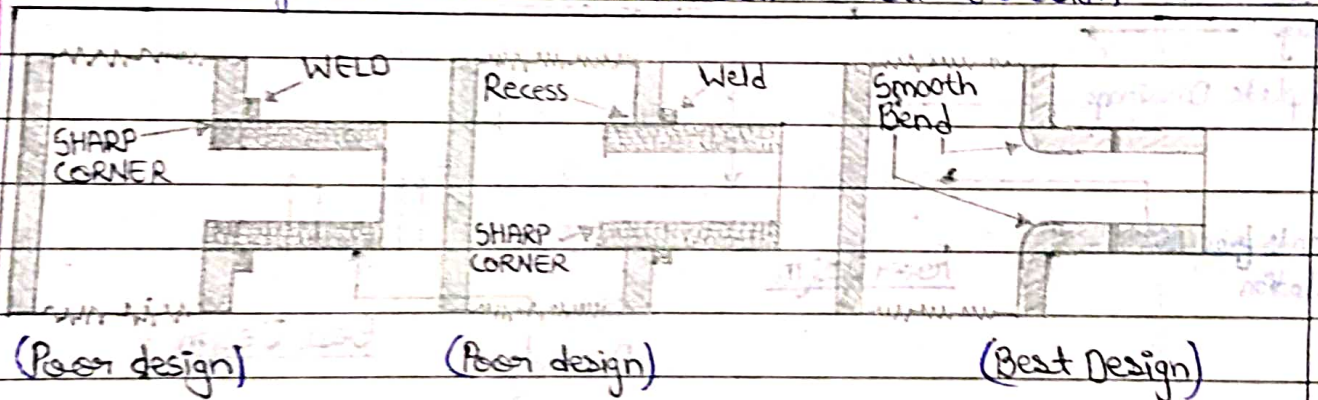
At cathode: $\frac{1}{2}\text{O}_2 + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 2(\text{OH}^-)$

Pitting is common in aluminium alloys, copper alloys, stainless steels, some of Ni alloys in solution containing chlorides.

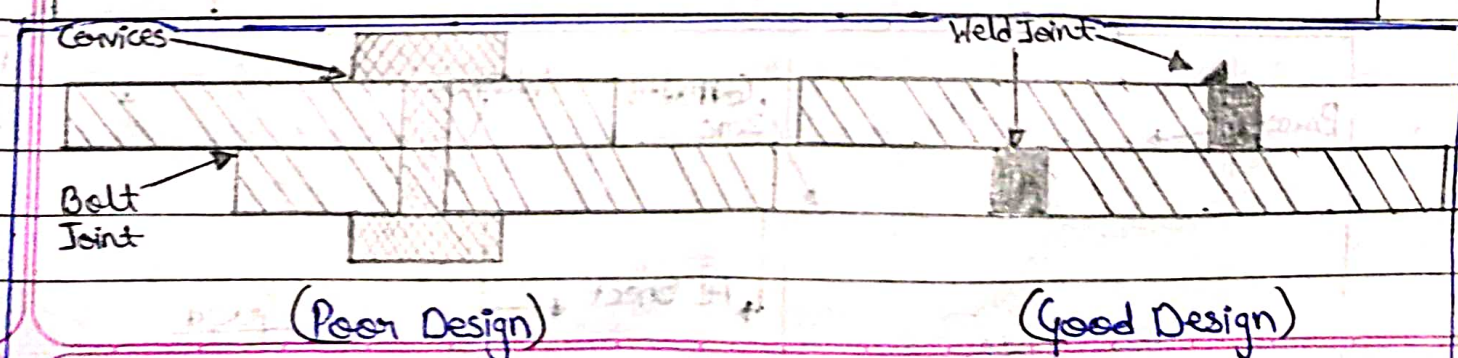
Protection Methods against corrosion

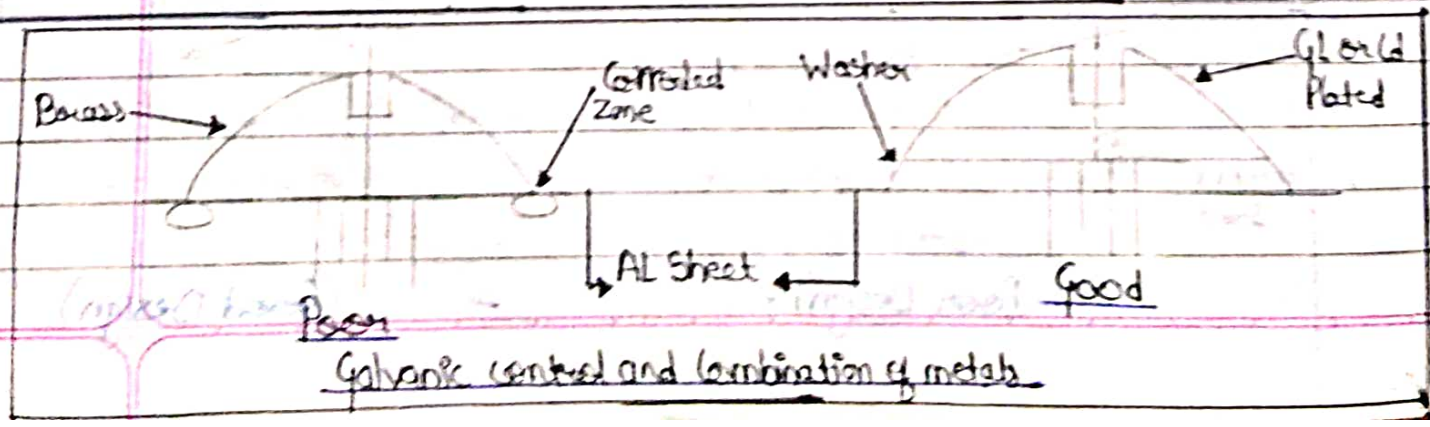
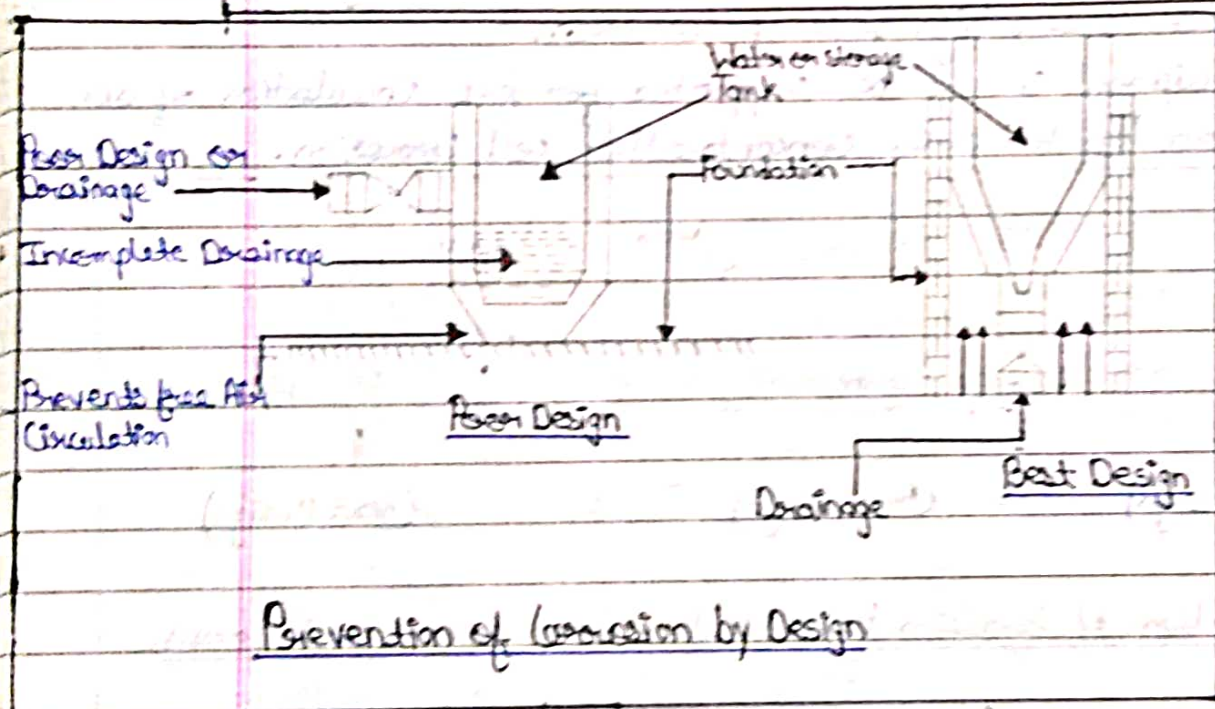
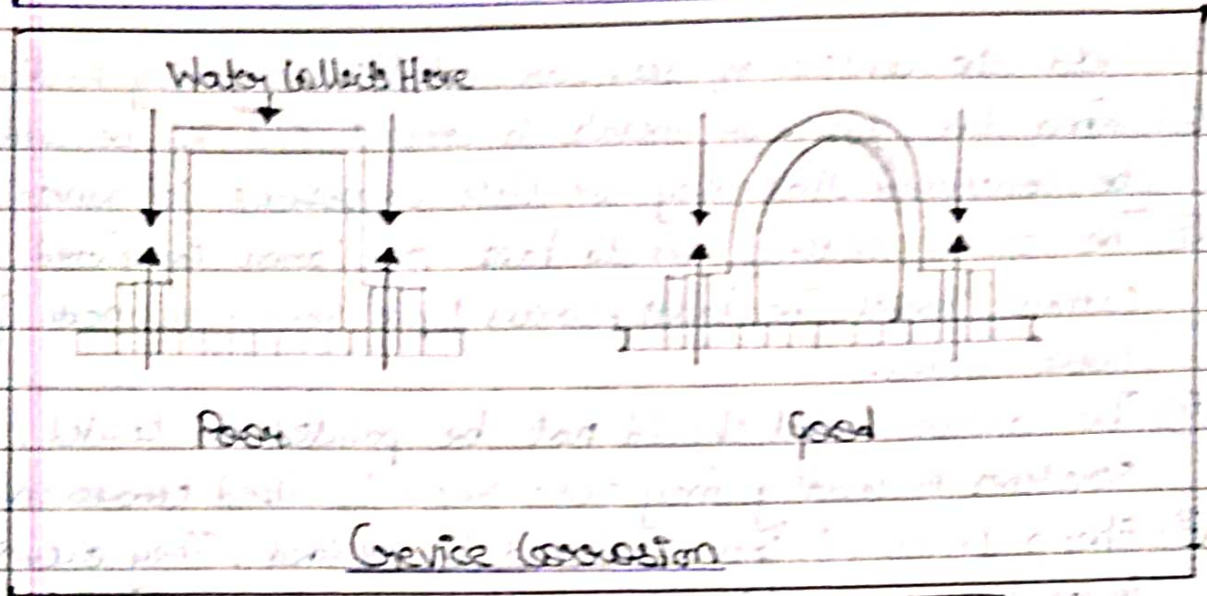
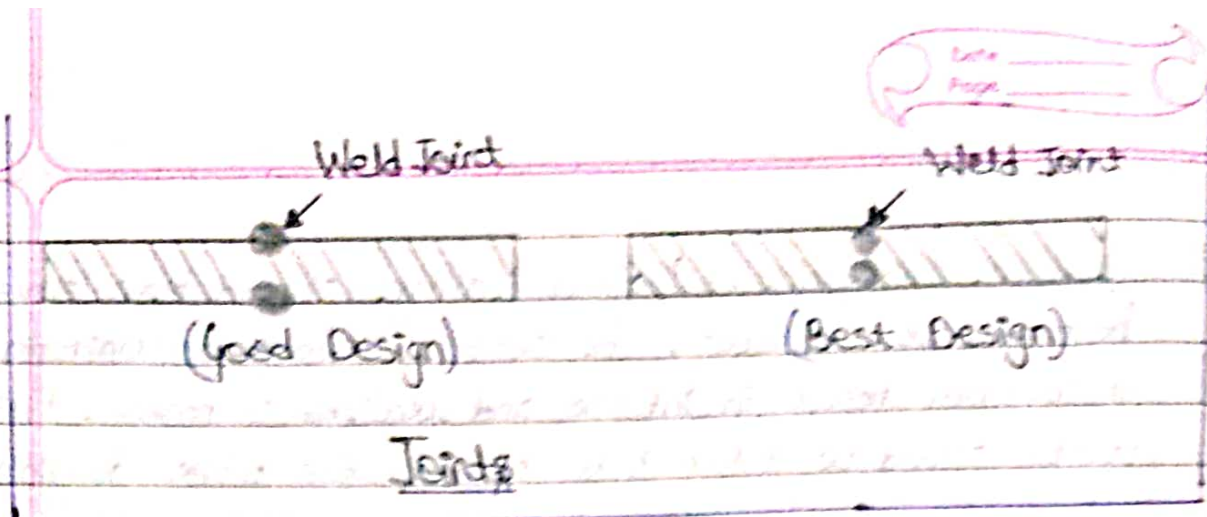
Some of the corrosion protection methods are given below:

- ① Proper Designing → Metals & alloys should have smooth & homogeneous surfaces. The surfaces should be uniform so that it will not result in intense and localized corrosion. Following points should be taken into consideration while designing.
 - ① Avoid the contact of dissimilar metal in corroding medium.
 - ② When two dissimilar metals in contact are to be used, it should be considered that they are close as possible in galvanic series.
 - ③ The anodic material should have large area compared to the cathodic metal. The bolts, screws & nuts should be made from more noble metal.
 - ④ The anodic metal should not be painted or coated. Any scratches in coating may cause rapid localized corrosion.
 - ⑤ Sharp corners & Grooves should be avoided. They accumulate impurities causing high rate of corrosion.
 - ⑥ The equipment should be supported for free circulation of air so as to avoid concentration cell corrosion.



Prevention of corrosion by elimination sharp corners and stresses.





② Selection of Metal $\Rightarrow$ Pure metal should be preferred since they are corrosion resistant. An alloy can also be used to avoid corrosion. Certain alloys are more resistant to corrosion than pure metal because it is homogeneous in nature.

③ Modifying the Environment $\Rightarrow$ The corrosive nature of environment can be reduced by:-

- (i) Deaeration (removes gases like O_2 , CO_2 , water, etc.)
- (ii) Deactivation, by adding chemicals which remove the O_2 for eg. Addition of Na_2S , Na_2SO_3 or hydrazine removes O_2 .
- (iii) Neutralization of acids by adding bases and Vice versa.

④ Inhibitors $\Rightarrow$ A substance which when added to the corrosive medium, effectively decreases the corrosion of a metal is known as corrosion inhibitor.

⑥ Anodic inhibitors get absorbed on anodic surface forming a passive film on it, (eg. chromates, phosphates etc.) thereby decreasing the corrosion rate.

⑦ Cathodic inhibitors slow down the reactions taking place at cathode. (i.e. H_2 evolution) eg. Organic inhibitors like amines, substituted ureas, heavy metals soaps etc. decrease H_2 evolution process.

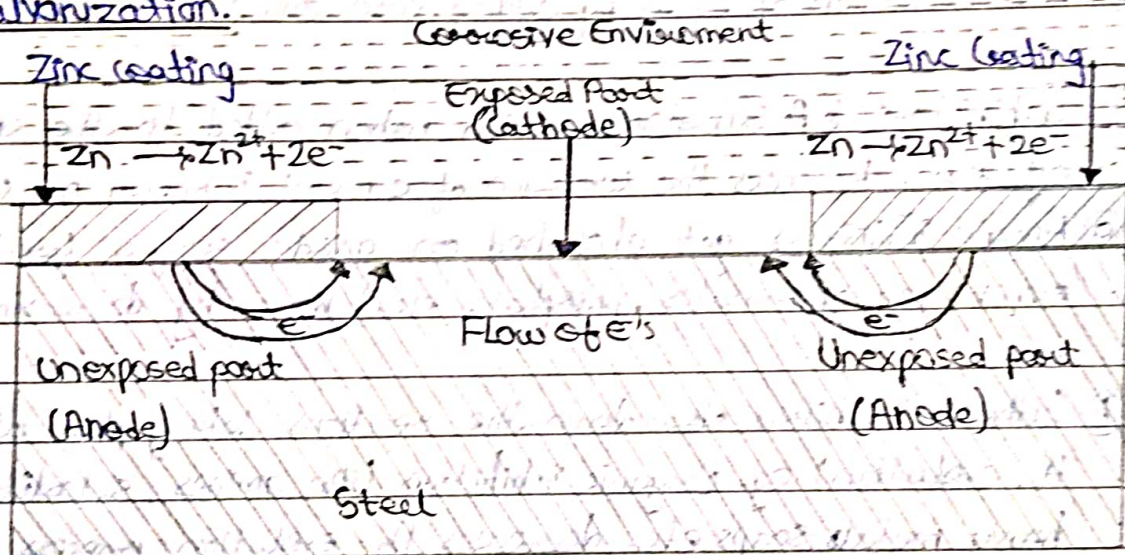
Mg, Ni or Zn salts deposits on cathode helps in decreasing the O_2 absorption process.

⑤ Protective Metallic Coatings $\Rightarrow$ Steel & iron, due to their comparatively low cost & availability, are commonly used in construction process. The protection of these metals are done by separating it from corrosive environment using different metallic coatings on it. These coating can be made by processes like electroplating, spraying, hot dipping etc.

Metallic coatings are of two types:

- (i) Anodic coatings, (ii) Cathodic coatings.

- ① Anodic Metallic Coatings = These coatings metals are anodic to the base metal which is to be protected. eg. Coating of Zn, Al and Cd on steel. Zn, Al and Cd are anodic to steel. If any breaks or pits occur on this anodic metallic coating, a galvanic cell formation takes place b/w the metallic coating & the base metal (cathodic), which is to be protected. The anodic metallic coating corrodes sacrificially & base metal being cathodic is protected. The rate of corrosion is also lower comparatively since area of anode is larger compared to the exposed area of cathode. Coating of Zn on Fe is known as Galvanization.



Functioning of anodic coating (galvanization)

- ② Cathodic Coatings = In this case the metallic coatings are cathodic to the base metal which is to be protected. Coating of Sn on Fe is known as Tinning. Here, Sn is cathodic to Fe.

On such coatings if there are scratches or cracks then it causes problem. The rate of corrosion becomes very high because, the area of cathode (Sn) is large here and the exposed area of iron (anode) is small. So such coatings are effective only when the coating is continuous and free from breaks or pits.



Functioning of cathodic coating - Tin plated on steel (Tinning)

⑥ Cathodic Protection Methods ⇒

This method involves, protection of metal by converting it into cathode. There are two methods.

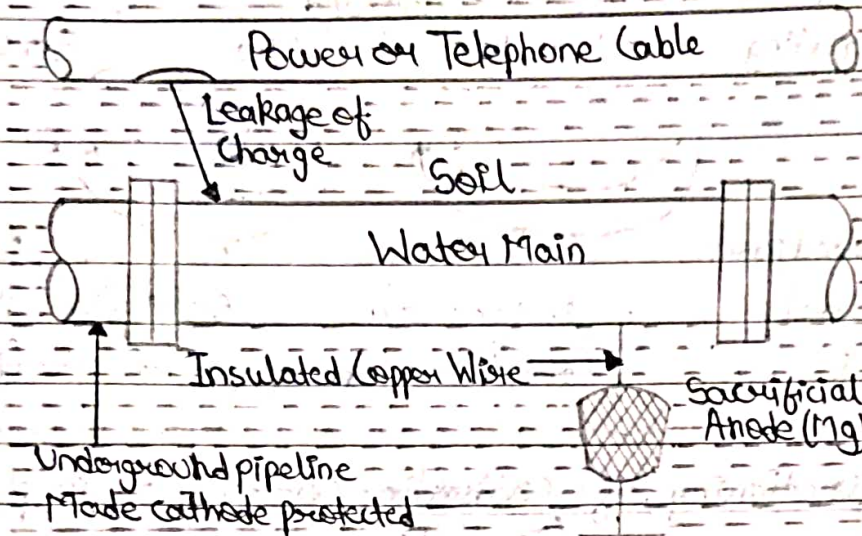
- ① Sacrificial anodic protection method.
- ② Impressed current protection method.

① Sacrificial Anodic Protection Method ⇒ In this method, the metal to be protected is converted into cathode by connecting it to a more reactive metal can act as anode. After a certain period the anode sacrifices itself and save the cathodic metal.
eg. Mg and Zn are being anodic to iron sacrifices to save iron when Fe is connected to Mg/Zn. Since Mg and Zinc sacrifices to save iron, they are known as sacrificial anode. Mg, Zn are Sacrificial anodes.

This type of protection method is used in industry, for ship hulls, buried water and gas pipelines, and even in railway track.

Ground Level

Soil

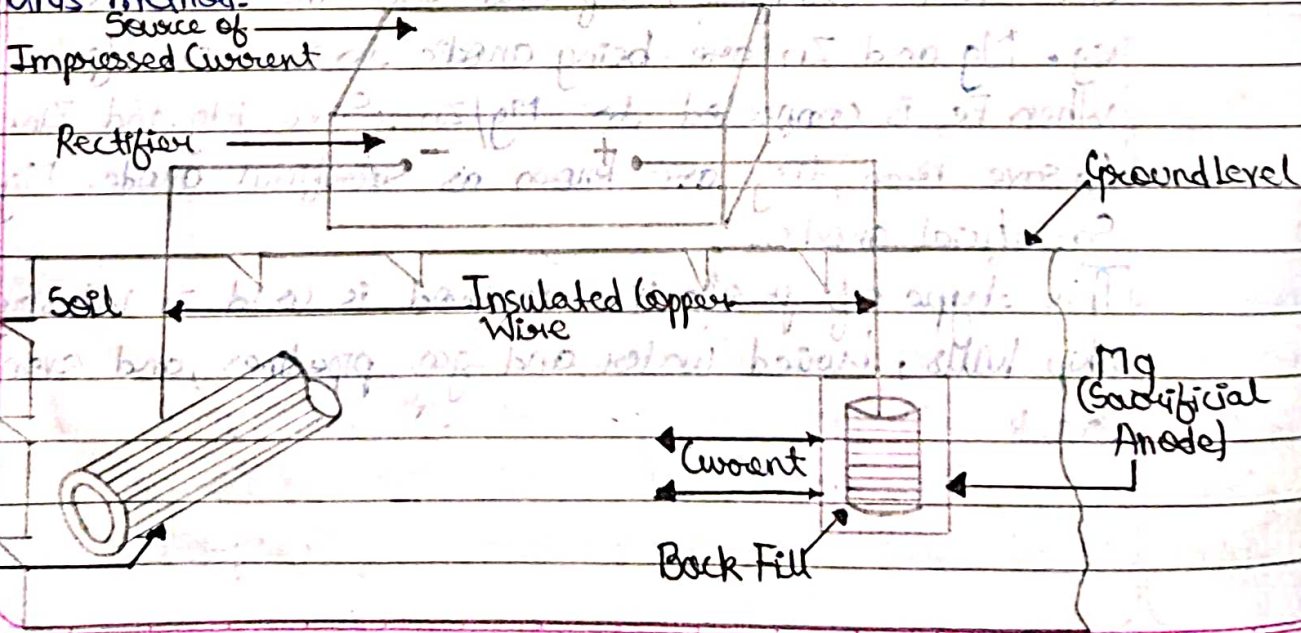


Cathode protection

(ii) Impressed Current protection Method - In this method, electrons are passed to the metal to be protected through a d.c supply and the metal is converted to Cathode.

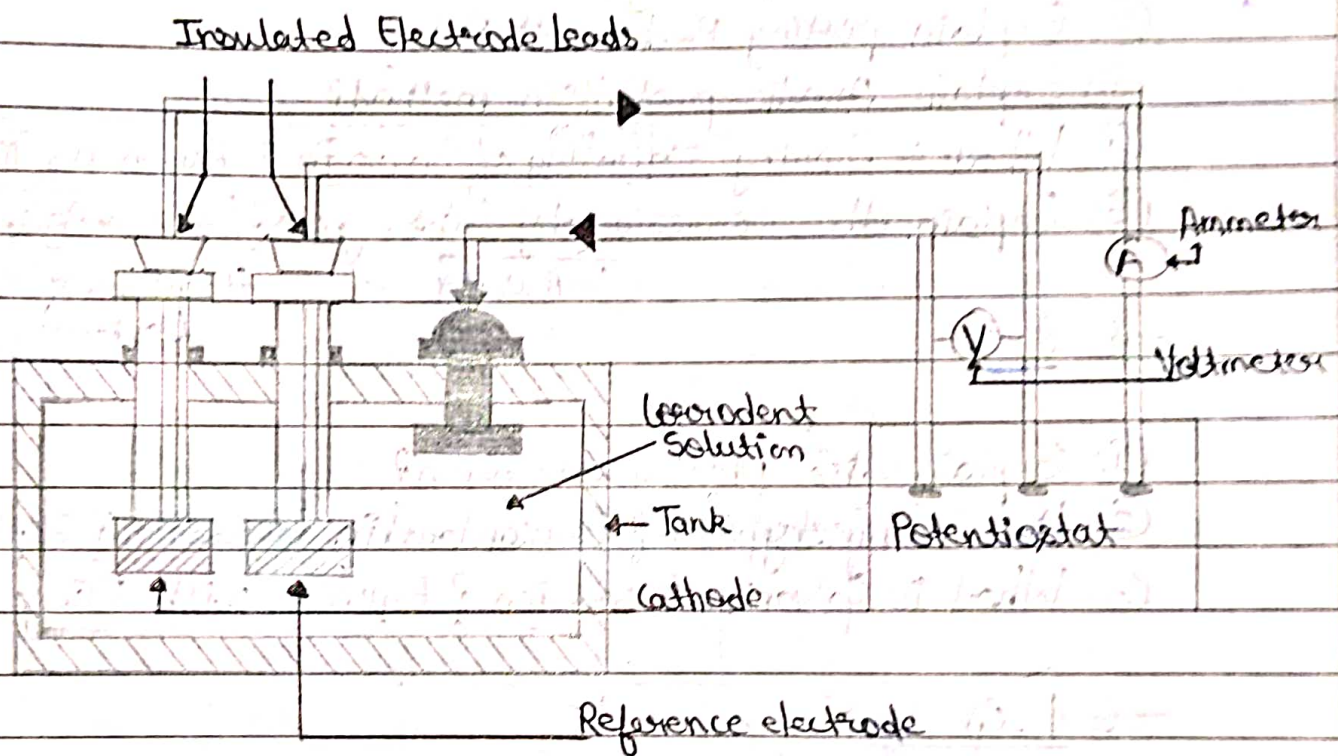
This method is expensive compared to a sacrificial anodic method. Anode is not necessarily be consumed.

eg: graphite or Pt used as anode for this purpose. This method is useful in protection of large structures which are used for long term operation. Protection underground pipelines carrying water can be done by this method.



Cathodic protection by impressed current

- ⑦ Anodic Protection Method :- By an externally impressed anodic current a protective passive film is made on the metal surface. Metals like Fe, Cr, Ti, their alloys etc. can be passivated by application of controlled anodic currents & the corrosion rate is decreased.



Anodic protection system for tanks.

→ Model question paper for Unit-2

→ V.S.A.Q

- ① What is the passivity?
- ② Explain pitting and Luggin's Rule?
- ③ Explain Anodic protection method?
- ④ What is Tinning? ⇒ Coating of Sn on Fe is known as Tinning
- ⑤ Explain the corrosion by other gases? ⇒ Cu & Ag (Ag in presence of H_2S in presence of Fe) form FeS (Non-protective, Porous).
(SO_2, H_2, Cl_2, H_2S)

→ S.Q

- ① Explain the pitting corrosion?
- ② How many types of Concentration corrosion? Explain in brief.
- ③ What is Galvanic corrosion? Explain with Diagram.

→ L.Q

- ① What is the Dry corrosion?
- ② What is Electrochemical corrosion?
- ③ Write about cathodic protection method?

V.S.A

① ⇒ Passivity is the phenomenon when metal & alloys shows resistive behaviour ~~to~~ for corrosion because of the formation of protective thin layer ^{on metal surface}. Which is insoluble, porous non-porous and have special ability which is self healing ability. It means whenever metal ~~to~~ will form deformities of broken or cavities it will self recovered or repair itself.

② Volume of oxide layer > Volume of metal ⇒ oxide layer is protective & non-porous
Volume of oxide layer < Volume of metal ⇒ oxide layer is non-protective & porous.